

# DR. RAYMOND CHRISTOPHER AMADOR

## PERSONAL DATA

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**Nationality** United States  
**Gender** Male  
 [Google Scholar](#)

## LANGUAGES

English (native), German (spoken and written fluency), Spanish (B1), Italian (A2)

## EDUCATION

DOCTOR OF SCIENCES: ETH Zurich, Zurich, Switzerland 3/2021 – 5/2025

- ▷ **Dissertation:** *Enhanced sampling of chiral reactions on chiral surfaces*
- ▷ Laboratory: nanotech@surfaces, Empa, Dübendorf, Switzerland
- ▷ Supervision: Prof. Dr. Daniele Passerone, Prof. Dr. Mathieu Luisier

MASTER OF SCIENCE: Humboldt-Universität zu Berlin, Berlin, Germany 3/2017 – 11/2020

- ▷ **Master's thesis:** *The electronic structure and optical properties of  $\text{Na}_2\text{KSb}$  and  $\text{NaK}_2\text{Sb}$  for photocathode applications: an ab initio study*
- ▷ Supervision: Prof. Dr. Caterina Cocchi, Prof. Dr. Thorsten Kamps

ERASMUS: Universidad de Salamanca, Salamanca, Spain 8/2015 – 3/2016

- ▷ **Research paper:** *Amistad Olvidada: el rol de España en la Revolución Estadounidense* (engl.: the Roll of Spain in the US War of Independence)

BACHELOR OF SCIENCE: University of Miami, Coral Gables, Florida, USA 8/2009 – 5/2013

- ▷ **Bachelor's thesis:** *The DXL Sounding Rocket Mission to Study Solar Wind Charge Exchange*
- ▷ Supervision: Prof. Dr. Massimiliano Galeazzi

## ACADEMIC & PROFESSIONAL EXPERIENCE

PhD Researcher 3/2021 – 5/2025

**ETH Zurich & Empa**, Dübendorf, Switzerland

Supervisors: Prof. Dr. Daniele Passerone, Prof. Dr. Mathieu Luisier

- ▷ Development and application of enhanced sampling techniques (metadynamics) and machine-learned interatomic potentials (using DeePMD and TensorFlow) to study chiral reactions on chiral surfaces, including the design of a custom chiral collective variable investigation of adsorption energetics
- ▷ Large-scale *ab initio* and classical molecular dynamics simulations (CP2K, LAMMPS, PLUMED), including trajectory generation, analysis, and validation against experimental observables
- ▷ Analysis of high-dimensional simulation data to extract reaction mechanisms, free energy surfaces, and structure–property relationships; implementation of automated Python-based workflows for reproducible data processing
- ▷ Collaboration with experimental groups and contribution to interdisciplinary projects in surface science and catalysis; supervision of Master's students and coordination of research tasks
- ▷ Conference contributions including DPG Spring Meeting (2023 Regensburg, 2024 Berlin), Surface Science Symposium (2023 Courmayeur), and co-organization of the MARVEL Junior Retreat (2024 St. Moritz)
- ▷ **Skills:** density-functional theory (DFT), molecular dynamics (MD), metadynamics, statistical analysis, scientific computing (Python, NumPy, SciPy), high-performance computing (HPC), data analysis pipelines, Git, scientific writing and presentation

Visiting Researcher

5/2024 – 10/2024

**Istituto Italiano di Tecnologia**, Genoa, Italy

Supervisors: Dr. Umberto Raucci, Dr. Peilin Kang, Dr. Enrico Trizio

- ▷ Development of machine-learned interatomic potentials (MLIPs) trained on trajectories from *ab initio* molecular dynamics to study chemical reactions on catalytic surfaces
- ▷ Used CP2K and LAMMPS to obtain the trajectories, and DeePMD to generate the MLIPs
- ▷ **Skills:** proposal writing (for computational time), Python (and libraries), Bash scripting, make files, GitHub, Git, high-performance computing architecture, GPU acceleration, pesto making

Graduate Teaching Assistantship

Spring 2021, 2022, 2023

Molecular and Materials Modelling: ETH Zurich, Zurich, Switzerland

**Level:** Master's, PhD

- ▷ Description: Python- and HPC-based implementation of concepts in the simulation of materials

- ▷ Content: HPC clusters; GitHub; Lennard-Jones fluids; theory of dispersion and van der Waals interactions; mean-field theories (density-functional theory, Hartree-Fock); molecular orbital theory; molecular dynamics; metadynamics
- ▷ **Responsibilities:** design, implementation, and grading of Jupyter notebook assignments; delivery and explanation of subject matter during lecture; administration of end-of-semester oral examinations
- ▷ Workload: approx. 10-15 additional hours per week during the semester
- ▷ **Skills:** instruction and supervision, didactics, syllabus planning, exam design, Git incorporation, exercise conceptualization

Student Assistant

11/2019 – 12/2020

**Helmholtz-Zentrum Berlin**, Adlershof, Germany

Supervisor: Prof. Dr. Thorsten Kamps

- ▷ Investigation of the experimental growth and characterisation of Cs-Na-K-Sb photocathode layers using the in-house growth apparatus

Scientific Assistant

9/2018 – 12/2020

**IRIS Adlershof**, Institut für Physik, HU-Berlin, Berlin, Germany

Supervisor: Prof. Dr. Caterina Cocchi

- ▷ Characterisation of the electronic structure and optical properties of Na<sub>2</sub>KSb and NaK<sub>2</sub>Sb for their use as electron sources in photocathodes for particle accelerators

Laboratory Assistant: X-Ray Physics

9/2017 – 03/2018

**Leibniz Institute for Crystal Growth**, Adlershof, Germany

Supervisor: Prof. Dr. Martin Schmidbauer

- ▷ Researching the scattering of incident x-rays at crystal surfaces to study electronic structure

STUDENT

██████████; Master, ETH Zürich

9/2023 – 2/2024

SUPERVISION

**Master's thesis:** *Modeling chiral reactions with ab initio simulations: anticipating future experiments*

- ▷ Currently: PhD Student with Prof. Dr. Mariana Rossi; Max Planck Institute for the Structure and Dynamics of Matter, Hamburg, Germany

██████████; Master, ETH Zürich

9/2024 – 2/2025

**Semester project** *Active learning a classical potential for a hydrogenation reaction on chiral surfaces*

- ▷ Currently: continuing student at ETH Zürich

MISC. PROF.  
EXPERIENCE  
(SELECTED)

Cocktail Bartender

3/2018 – 2/2021

**Süß War Gestern**, Berlin – Friedrichshain, Germany

Barista and Bartender

7/2016 – 2/2021

**The Caramina: Persian Kitchen**, Berlin – Neukölln, Germany

Extracurricular Activities Advisor

12/2016 – 8/2019

**Humboldt-Institut**, Berlin – Mitte

Teacher of English

4/2014 – 12/2014

**University of Cambridge – CELTA**, International House, Seville, Spain

REFERENCES



AFFIRMATION

**TBILISI** 🇧🇪. Saturday 18<sup>th</sup> July, 2026